ActInf Livestream #009.0: "The Projective Consciousness Model and Phen

Livestream | 1:48:16

hello and welcome everyone to the active
inference live stream I believe we are
live it is ACM stream
9.0 and it is November 18th
2020 I am Daniel fredman and I will be
doing a solo active stream today so
welcome to teamc com everyone we are an
experiment in online team communication
learning and practice related to active
inference you can find us on Twitter at
inference active active inference
gmail.com our public keybase team at our
YouTube channel or um through any other
mechanism this is a recorded and an
archived live stream so please provide
us with feedback so that we can improve
on our work all backgrounds and
perspectives are welcome here and for
video etiquette on live streams make
sure to mute if there's noise in the
background raise your hand so that we
can hear from everyone on the stack and
use respectful speech Behavior so just a
quick announcement before we jump into
the paper discussion you can find on our
Twitter information about the rest of
our discussions for 2020 we have six
more uh group act imp streams in
2020 on November 24th and on December
1st we're going to be in act imp 9
discussing the projective Consciousness
model and phenomenal selfhood and that
will be the paper that this actm stream
9.0 is context
for in active 10 on December 8th and

15th we'll talk about the paper a variational approach to scripts while in actm 11 on sophisticated affective inference we'll talk about more modeling and math again and that will be on December 22nd and 29th so that will close out 2020 and then we look forward to a whole new panel of events and projects collaborators for 2021 so find out more on our Twitter that way let's talk about what we're going to do in actim stream 9.0 which is right now so the goal of actim 9.0 is going to be to set the context for 9.1 and 9.2 which are going to be the group discussions that we have on these papers and the paper that we're going to be discussing for 9.1 and 9.2 is called the projective Consciousness model and phenomenal selfhood by Will aord at all
2018 and there's a link that people can see the video that we're about to jump into right now is an introduction to the context of some of the ideas presented by the authors it's not a review or a final word it will contextualize some of the Ideas Math and vocabulary that might be interesting or important to know for reading will afford at all 2018
and the punchline that they're going to work their way towards is that we might be able to combine Fields such as geometry cybernetics and neurophenomenology using the framework of active inference and the free energy principle to potentially understand or explain Consciousness so that's going to be up to you up to us how we feel about

these kinds of claims that they're going
to get to in the paper but that's where
we're headed and the sections of 9.0
will be as
follows first we'll go through the
keywords and hopefully provide some
background on the different keywords
whether it's your first time hearing one
of these words or whether you're in one
of these fields and then we'll talk
about the aims and claims of the paper
go through their abstract and the road
map the sections then we'll talk a
little bit about Consciousness from the
author's perspective just what do the
authors say is important about
Consciousness and what do they think are
invariant aspects of it we'll then walk
briefly through the figures just try to
understand what are these figures even
doing how are we going to get a first
pass on understanding what these figures
are all about and then in 9.1 and in 9.2
we're going to stay with this exact same
paper so save and submit your questions
and feel free to get in touch with us if
you'd like to participate whether that's
in
9.1.2 or in a future Endeavor let's talk
about the
keywords the first keyword is uh
actually not on the paper keywords but
is what we're about here is active
inference and the free energy principle
that's what we'll try to tie so much of
this back to and here where the keywords
that were listed cybernetics projective
geometry Consciousness

neurophenomenology firsters perspective
and perspectival Imagination so cool
terms maybe it's your first time
thinking about math and geometry in the
context of Consciousness maybe it's not
but hopefully as I go through these
following keywords I hope that this is
interesting and exciting whatever level
of familiarity you have with these
Concepts so first keyword is cybernetics
and it's also written there in Russian
as kuberi and um there's many resources
to read on for the history of
cybernetics some of which we're going to
talk about here because in a lot of
other act dstreams and other locations
you can probably hear about the
mathematics underlying cybernetics and
about how it's related to control both
things that we've talked about on this
stream but let's talk a little bit about
just more historically or from a um
cultural perspective what is cybernetics
what is it trying to do cybernetics is a
school of thought tradition of action
however you want to think about it
that's related to the action policy
selection and control of goal oriented
systems now there's various orders of
cybernetics which we can sort of walk
through here and um I'll go through a
couple of different resources on
cybernetics and hopefully you'll see
soon why I put a history citation on the
bottom left and a little bit of Russian
on the top and it's not just because we
have great Russian colleagues in team
comp here are the orders of cybernetics

there's probably different ways to State them or think about them first order cybernetics might be thought of as self-regulation without detailed observations so this could be like a negative feedback system that is able to maintain a semblance of homeostasis in virtue of being an isolated or a closed process like there's something that gets triggered when it's too hot and then as soon as it's triggered that it's too hot it instantly kicks in a cooling program second order cybernetics is um self-regulation that can be influenced by observers within a domain of control so that could be for example a continuously varied um thermometer that's able to enact different kinds of control policies as a function of continuous inputs from the thermometer third order cybernetics is self-regulation that is is influenced by observation with a relational basis or an ecological grounding so the system in this third order of cybernetic Loop is able to acknowledge its surroundings and learn develop and evolve and fourth order cybernetics is when self-regulation involves the enactment of multiple different realities or adjacencies affordances counterfactuals a lot of ways you can think about that in uh terms of a functioning creative Observer so these are deeply contextual utiliz and Integrated Systems that are doing higher orders of cybernetics and I'm not a stickler about which order is which

number I also on purpose drew the
parallel with the um one two and three
Loop thinking which is related to um
some decision-making Theory from
Innovation and the entrepreneurship
space but I just thought it's
interesting here's a figure from the um
book chapter cybernetics in the 20th
century on the bottom left there and it
shows a map of Concepts and how the
related to cybernetics so on the left
are sort of two inputs to cybernetics
which are control theory and the idea of
control and then communication Theory
and the idea of communication and
information and so those are two
branches that we've talked a lot about
on this stream and we will continue to
talking be talking a lot about because
control of action and communication
message passing these kinds of things
inference is the thread that runs
through them all these kinds of things
are critical for control theory for
cybernetics for the kinds of systems we
want to get into working with on the
right side we see cybernetics branching
back out into the machine area with the
engineering sciences and the planning of
for example complicated Factory
operations but as well as into the
animal and the social levels so the
animal cybernetics is about for example
the self-regulation of behavior and the
social cybernetics could be about um the
way that our governance is run and
there's a really interesting book read
plenty that's kind of like historical

fiction about cybernetics speaking of History this is such an interesting plot and it was in the cybernetics book figure six it is the usage of the term cybernetics and control by years in the text of populations Publications indexed by Google Scholar so caveats maybe people are speaking about it with different words or it's being used uh different language not English or there's some other way of saying it but look at how similar these things are in terms of their Trends where both of them ramp up gradually leading to in early 2000s a lot of t discussion of cybernetics and of control potentially looking backwards a couple years so think about these peaks in the early 2000s as potentially reflecting something at the close of the 1900s and then how they both start to drop off and this is consistent with something that a lot of us have experienced which is that cybernetics is a term that a lot of at least English speakers may not be too familiar with but we can look at the journals that friston and others have published in we find a lot of connections to
cybernetics what is going on there so to get a little sense of what is going on two other resources to look into would be this big history understanding of complexity information and cybernetics and then this book which I'll read a quote from right now which is called from new speak to cyber speak a history of Soviet cybernetics by Slava gich and

the author writes in this book from 2002
the history of Soviet cybernetics
followed a curious Arc in the 1950s it
was labeled a reactionary pseudo science
and a weapon of imperialist ideology
with the arrival of kev's political thaw
however it was seen as an innocent
victim of political oppression and it
evolved from a movement for radical
reform of the stalinist system of
science pretty interesting in the early
1960s it was hailed as a science in
service it was hailed as a science in
service of Communism but by the end of
the decade it had turned into a shallow
fashionable Trend using extensive new
archival materials gerovich argues that
these fluctuating attitudes reflected
profound changes in scientific language
and research methodology across
disciplines in power relationships
within the scientific community and in
the political role of scientists and
engineers in Soviet Society his detailed
analysis of scientific discourse shows
how the new speak of the late stalinist
period and the Cyber speak that
challenged it eventually blended into
cyber new speak so maybe there was a a
book summary of this um work but pretty
interesting how that plays out and so
definitely we'd love to learn more about
this whether uh we have Russian speaking
colleagues who could fill us in or
anyone else next keep word projective
geometry what is projective geometry
about well of course you could go down
the rabbit hole with this topic but

let's just try to keep it constrained to
what the authors are going to use it for
in terms of the paper they write the
concept of a 4D four-dimensional
projective transformation is Central to
the model as it yields an account of the
link between perception imagination and
multipoint of view action planning great
okay so what is happening here and how
can we understand this and its
relationship to projective geometry I'm
going to introduce just two ideas here
with this visualization the first by
looking at the bottom is the distinction
between the ukian and the non- ukian
types of spaces and then the second idea
with a purple arrow is this idea of a
map or potentially you could call it a
mapping because it's actually a process
and it's actually a certain type of
process it's a projection so when we're
talking about projective geometry we're
talking about geometry which means the
The Shape of Things and the spaces that
they're existing in so as opposed to
topology which is how things are conned
connected with nodes and edges geometry
is about things and their shapes and
their spaces we're going to talk about
projections amongst different geometries
specifically potentially this one
between ukian and non- ukian geometries
and here's a quote that they write
authors do they write this perspectival
phenomenal space so the space in which
they're going to say that experience
occurs within phenomenology occurs
within a space allows us to locate

ourselves and navigate in an ambient space that is nevertheless essentially ukian however in a ukian space there are no privilege points no points at Infinity no Horizon and no and an arbitrary origin they say there's no nonarbitrary origin same thing this very this very strongly suggests that a non- ukian frame must be operating in the phenomenal space with the help of sense motor calibrations to preserve the sense of the invariance of objects across the multiple perspectives adopted and relate the situated organism to the ambient ukian space so what is it that allows me to continue thinking about this as a pen even as it comes very close to my face and looks very weird so what is that generative model that allows me to act like everything's normal as objects maintain their invariances as the eye travels through the space and so we can contrast this ukian space that they're supposing that we exist in they're saying we exist in a space that doesn't have privileged points of access no points at Infinity etc etc um and they're saying the space that an organism for example moves through is ukan while the space that their Consciousness is being projected into experientially is non- ukian what about it is non- ukian well let's just take the opposite of what's in red and put it in blue so there are privileged points from the point of view of the Observer that is the privilege point of view quite literally there can be points at

Infinity there can be points that sort
of just like they're so far away they're
infinitely far away and not just from an
affordance point of view but from a
perception point of view there can be a
horizon in fact we see it every day and
there can also be a non-arbitrary origin
which is a little bit of a restatement
the idea that there could be a
privileged point but basically it's
privileged it's a non-arbitrary origin
these are pretty similar Concepts now
non- ukian it's like saying nonlinear
and the joke about this is things that
are not elephants that's the non-
elephant class you'll know it when you
see it so that's about how helpful it is
to say it's nonukan or it's nonlinear
model okay we get it it's not linear
model it's not ukian geometry but
there's many choices and the reason why
even though it sounds like wait you're
barely ruling anything out we're
actually ruling out very specific parts
of the geometry of the ukian space so in
the ukian space parallel lines don't
intersect but that's not necessarily um
the case for certain types of geometry
or a triangle with three edges and three
angles is 180° from the sum of the
angles but on a sphere it's not that way
so that is what is happening with these
geometric projections is we're going to
be projecting from sort of the um grid
they want to think about now you might
want to ask Bucky Fuller if we're
actually in a ukian space or if it just
seems that way because of public

education but for the purposes of this
paper we're in a ukian space as action
and that's the space we want to be
making our perception uh simulate so to
speak and to act like everything is
normal and to decrease our uncertainty
about a ukian mapped space while also
relating it to through a projection
something that does have a privileged
point of access and something that is as
they put it informed by sensory motor
calibration and relates to our action
affordances so that's how you can have a
ukian concept of for example um a gap in
the ground that you can't jump over so
it's a very complex estimation of what
you can do and how big the Gap is and
where you are how much runup you would
need Etc all of that is going to come
together by doing what they're going to
talk about here now Consciousness um
it's great to talk about Consciousness
and I'm going to go into a little bit of
detail soon about some of my own
thoughts on the issue but when talking
about Consciousness it's important to be
really clear even though it's something
that we're not always sure about It's
always important to at least use
language very carefully and use our
scholarship and our Intercultural
respectfulness very importantly because
as I'll hopefully come back to a few
points soon it's just a critical issue
and we can't think of it um just
flippantly the authors write when
talking about Consciousness because I
want to just go with what they think

about Consciousness for here they right we might divide contemporary neuroscientific and psychological theories of Consciousness into three main types the biofunctional the cognitive neurofunctional and the information
theoretic as should be clear the pro Ive Consciousness model the PCM incorporates the emphasis on self-awareness common to many approaches to Consciousness as well as an emphasis on the biological functions of Consciousness so they're defining the space a certain way I'm going to Define it a different way soon but they're defining it that way then we're going to insert the PCM here and then they write at the end the PCM thus subsumes and unifies the main insights in all of these approaches to Consciousness above all what we add to pre-existing Theory is a geometry the thesis that projective Transformations and projective frames necessarily subtend the appearance and workings of Consciousness so whatever we're going to learn in this ACM stream or whatever we think the PCM is doing and we want to evaluate it not by um what it doesn't do but what it does do and what it claims to do what it sets out to do that's where they're trying to step in their understanding of literature and Consciousness is that they broke Consciousness down into these three kinds which ation exist there you could read the paper and then they're going to step into that Gap and then they're

going to make the claim that they're subsuming and unifying the main insights from these approaches where does that leave these approaches from a theory perspective question for another day but that is where the authors understand their work to be fitting in so if you don't know what the PCM is or it's kind of trippy to think about what it means to do mathematics and geometry of Consciousness that's perfect here is the blank that they're going to fill in with their work what what do I think about Consciousness though and I want to be clear about what my stance is or at least just give the tip of the iceberg on my stance because this is a first person actim stream and uh there's some relevant research I'll discuss in a few minutes and also I've just for a long time been interested in the science of Consciousness but also this second layer of How It's communicated especially from scientists so there's a lot of places to go here and a lot of places to start I'm just going to walk through a few General thoughts and raise a few questions questions really more questions than answers and then I'll discuss a paper from the last couple of years um with a colleague Eric
sovic all right so there are many ways to think about studying Consciousness and one path that you could go down um this is sort of you know across cultures through the time One path you can go down is thought experimentation so this could look like meditation it could look

like a specific thought experiment um or
some other sort of insight approach or a
gankin approach and whether it's
culturally called one thing or another
is not as much my concern here um the
thought experiment approach doesn't
involve making interventions for example
in the outside world just thinking about
Consciousness you know hearing TED talks
and stuff like that so the issue is that
um whatever is actually happening if
that's what we're pursuing um we could
just be led to the wrong conclusion and
then if there is some element of free
will or interventionary role of
Consciousness then we could intervene to
um end up believing that couldn't
intervene or any of these combinations
so I just never quite understand the
relationship between what one directly
feels convinced about with respect to a
thought experiment or a thought or an
idea and how that might relate to how it
actually is for people who aren't that
person and one aspect of this is that
it's really impossible to know which
aspects of thought experiments can be
transferred to other situations so for
example one of the most famous papers in
this area most cited papers is a 1974
paper by a philosopher named Nagel and
that raises the question what is it like
to be a bat I don't think he invented
that question um but he's the one who
got famous for it and in that paper it's
written an organism has conscious mental
States if and only if there is something
that it is like to be that organism

something it is like for the organism
now you might
think what are philosophers doing
because it sounds like a sort of a weird
question and I understand that many uh
careers and a lot of good thinking has
come out of this question so I want to
optimistically see it as a catalyst in a
seed um it's always good to have simple
questions to return to it's good that
it's such a simple question and it opens
to so many perspectives it initiates
students into our knowledge Traditions
there's so many advantages to having
these kinds of classic works but to be
that um uh to be blunt I guess I found
the phrasing of this question of
Consciousness to be quite unhelpful
because it is so informal it's so
hypothetical it's untestable it's
ambiguous but it's specific it brings a
bat in to question and then you can
imagine oh is a dog what is it like to
be a dog now these are all good things
to imagine but again they don't really
bring us closer to addressing real
Frameworks for Consciousness and the
reality is there's been now a vast
Wasteland of literature that's just not
comparable to other work in the same
exact space across different species
that essentially amounts to speculation
and appealing to what it might be like
to be a certain type of animal thinking
well if I was really a bat and I grew up
as a bat would I ever like thought uh
what it was like to be a human how I so
it ties up to all these second level

questions that aren't super relevant and
um we've always wondered through time
every culture everyone has wondered what
other systems experience whether that's
a person or an animal or a fictional
character or a physical object so the
speculation and the think pieces they
don't really add something substantial
to the literature which is what the
scientific literature is about for me so
the bigger issue though with thought
experiments of any kind is that we can
always imagine things that are just not
making sense Within framework or any
framework or just don't relate to the
world that we live in um we all know
that we can be wrong about an idea but
believe that we're right and so um to
kind of rephrase that in an active
inference framework we could say that
these are situations where the Precision
parameter is very high there's a belief
a metacognitive belief that the
Precision is high that some question has
been resolved um this represents that um
for example someone's model of
Consciousness whether it came from a a
knowledge tradition or their own
thinking they believe they're they're
onto it they got it figured out it's
resolved for them but um that doesn't
mean that your s your estimate is
related to S hat the actual state in the
world and I'm not passing judgment on
being overconfident or um being wrong
because sometimes it can be very
important and Powerful to have these
kinds of beliefs but that's not what

being correct is about and so if you
actually don't care about being correct
being right going towards the truth and
you really don't care about that then um
I'd love to hear where you're coming
from
um the rest of the scientific literature
although with many shades and many
differences is moving towards what
actually would be General generalizable
through truth so let's go back to
consider experimental approaches which
is something that comes up a ton in the
philosophy and the science of
Consciousness literature now it would be
awesome to do experiments on
Consciousness and in some way don't we
all we could study humans as they behave
and sleep and daydream take um
psychedelics or they wake up for
anesthesia and they develop and there's
so many things that could be exciting to
learn about also if we can do
experiments on Consciousness we could do
experiments on animals now we could do
experiments that are interventive like
um manipulating the brain of an animal
or the body or the situation of an
animal these experiments might be seen
as inhumane or just to massively violate
ethical standards for humans and so
there's a bit of irony in doing these
potentially invasive non-consensual
experiments with captive or wild animals
when we're studying potentially their
capacity to not consent to our
discussions with them um it's a
interesting question I won't go down the

ethics uh route here but a non-ethical
question that's related to
experimentation is that we just don't
know which systems to study or what
observables to measure for experiments
and so we're studying something that
isn't really measurable or observable
but we'll come back to this idea of
observability in a few minutes when
we're talking about
neurophenomenology and the reality is no
matter what you do measure even if you
come up with this one special
measurement that hasn't been done before
all experiments are interpreted within a
cultural and a scientific milu in terms
of what measurements conclusions
interpretations are ethical correct
accepted valid discussed all these
things and even going another layer back
about experiments even assuming that we
can measure what we want to know and
measure and everything experiments um
especially if they're disconnected from
grounding in theory it's difficult to
know what is really being uniquely
tested or demonstrated or falsified by
different kinds of experiments or what
we can learn from different kinds of
measurements and um there's a lot of
ambiguity a lot of levels the ambiguity
can enter into but just to give a few
examples an experimental result could be
consistent with many different
Frameworks or potentially all Frameworks
or a set of Frameworks for Consciousness
so when we really want to be identifying
what Frameworks we're wanting to work

with it doesn't narrow the space as much to do experiments which will return too soon and that's kind of related like the neural correlates of Consciousness debate which is do you put somebody in a neuro imager and then have them do a no report Paradigm where they don't say yep I heard the sound so you play sounds of different uh you know notes but what if it just subconsciously influence their brain there still could be a neural signature of that how would you know they experienced it well that's called a report Paradigm you ask them did you experience it but in some sense you're just asking them if they remember or if they hit the button so it gets messy but that really gets into some of the the Weeds about what you're measuring and another example or area where we see limitations on experiments related to Consciousness is an animal behavior so animal behavior seems naturally like it'd be adjacent to human consciousness studies and we see a wide variety of things being used here like pulley tests can two of the same animal pull a pulley so they get a reward looking in a mirror can one recognize themsel and take an action pattern to clean thems in the mirror and the first thing to note is that whatever kind of setup you're dealing with a visual illusion a cooperation task a recognition task memory planning they're always going to be based upon the affordances of that animal and again it might be ambiguous as to whether they're using some CU that

you actually expect um or if it doesn't
see it's not going to be able to be in
the mirror test so at the very least
these tests for Consciousness which
we'll get back to don't help us really
study the ground rock of this Frameworks
for Consciousness question because all
you could say is okay this bird can do
the pulley test but we don't even know
what that means and then of course if we
find out that potentially experience
isn't related as much as we thought to
behavior or one has a model that it's
not related to behavior then you can say
yeah it's a robot that just acts like
it's a crow all right I think I'm still
streaming but it looks like my camera
froze I'm going to just replug it in
um yeah
actually I'll just see if don't know
what it's about but that happened uh
last time as well but sayy I'll just
close the camera for
now
um actually let me just see if I can
remove the source and then add the
camera back in we're going to find out
live whether it is OBS being weird or
whether it is um the actual USB
connection
and turns out it was
OBS we now know and we're right back
in it
so um to the point about potentially
robots or zombies as denet might put it
doing uh complex behavioral tasks uh you
wouldn't know whether the experience was
being had even if the behavioral

experiment you had lined up for this
animal was enormously complex and uh
robots might be able to do things that
we might not expect them to be able to
do already such as uh learning language
or being culturally embedded and so yeah
another Point here is that before you
get too excited about any single
experiment that might prove or disprove
a model of
Consciousness consider it just another
grain of sand on a sand Heap and any
single grain of sand that you're adding
to this Heap now could be the one that
sets the whole Cascade in motion so by
all means I encourage experimentation
but that being said let's not fool
ourselves in pretending that many of the
data points we already have are out
there in the world already and that just
just one more experiment is unlikely to
provide any type of knockout evidence
for or against a model of Consciousness
and so to kind of um recap there there's
the thought experiment Branch the
philosophy meditation Insight route and
there's the experimentation
interpretation route and uh those are
connected of course experiments are
ideas too but that being said they both
have pros and cons and people will often
argue one or the other as if they have
um the answers from one branch or the
other I hope I've demonstrated that that
isn't true and
one reason I think that it's important
to have this Viewpoint is that we're
living in a world just as a prima fascia

observation where there's some people humans who say that they think all things are Consciousness or that all things are Consciousness in their own special way or that nothing is conscious including humans and of course there's many flavors to each of these viewpoints that could be held um or at least communicated and there's many things about beliefs about Consciousness that might not be verbal or expressible in a certain language so the point is we're in a total state where the evidence in the world has resulted in people believing different things through time different things even within the same family and so let's be humble the table has mostly been set for us in terms of observations about ourselves and about our world though it doesn't mean we can't reimagine much of our own experience or the world but um it would really have to be quite the thought experiment or quite the experiment uh in the real world if it can resolve these long-standing and philosophically grounded differences in worldview between schools of thought and between people and even if it did this even if the experiment were so good that it just convinced everyone it wouldn't even mean it was right it would just mean that it was convincing as a meme and so if we're really trying to find out what's right rather than how can we shape our own reality and other peoples which isn't always the worst task if we want to be right then as far as I know it's not

going to be the experiment or the
thought experiment route alone and one
last point on this slide because there's
a lot to say here is that there's
academics who believe different things
and you can see these slides and
citations and see that um they cite
different work that says one thing about
the global workspace or another thing
about information Theory there's also
different views on Consciousness from
different world Traditions from artists
from practitioners from non-scientists
from non-philosophers from all kinds of
people people children adults everyone
this is just to say that we should be
very cautious about any kind of
hegemonic especially scientific effort
to define consciousness whether it comes
from Academia or from some other source
there's just way too much on the table
here for people to be making sweeping
claims about what is or is not conscious
uh and there's a lot that is Downstream
to that it's really a personal and a
cultural thing what somebody believes
about Consciousness um what they want to
say or what they want to commit to and
um also while some people might find
these topics exciting I think they're
exciting if you're listening you
probably find them to some extent
exciting as well but many people are not
interested in considering it and don't
perhaps want to be told this is how
their Consciousness or experience is and
other people are simply okay not knowing
whether they're curious about it or not

and so I'm always skeptical about
science says type thinking which can be
often how scientific papers are written
even if it's written in an author's say
type thinking it's about how it's
communicated we've all seen the misuse
of science A's policy related to
everything and uh Consciousness would be
the next level on that wouldn't it be so
fill in the blanks think about what kind
of power is held by the people who can
say what or who is consciousness or
evaluate conscious states such as
suffering or pleasure so there's just a
lot to it and then um the last Point
really here is that people can change
their beliefs during their own Lifetime
and go deeper and shallower into
different ideas and that probably
doesn't change the underlying nature of
Consciousness so it will never be enough
how we think at one moment or for a
group of people to believe something
I've gone down these discussions and
Roads many times with people um alone
and in conversation and people will tell
me well you know we're working with our
best possible understanding of
Consciousness given the research at this
time and it's just simply not true
there's so much to know and respect in
terms of the research and also the
non-scientific literature on
Consciousness that there is just a lot
more to the discussion than well let's
just tell the kids something simple to
it now to go beyond that and thanks for
bearing with me it was just I wanted to

say a few of those things because a lot
of times Consciousness papers are
approached from
non-scientists with excitement so I want
to provide a little bit more of a
nuanced scientists take here on how we
can really respectfully study
Consciousness because this isn't just
like frisen at all figuring it out in a
paper but I love the paper and it's
interesting so how do we hold this yes
and perspective through Nuance hearing
multiple people's perspective having
participation so what I'm going to talk
about here is this paper that I worked
on with my colleague Eric sovic who's a
professor in Norway and the paper called
the ant colony test as a the ant colony
as a test for scientific theories of
Consciousness so this is how I came to
be thinking about a lot of these topics
now Professor sik's background was in
neurobehavior and in genetics of the
honeybee mine was in similar areas but
related to ants Eric is also a very
pragmatic and a philosophical thinker
his PhD adviser who is Andy Baron uh at
M University in Australia is a honeybee
biologist and Professor Baron has also
written philosophy papers that are
related to entomology as well as to
experience and Consciousness uh on my
side my PhD adviser Professor Deborah
Gordon at Stanford is an ant behavioral
ecologist who also wrote a lot of
philosophy Works broadly construed and
so both Eric and I had really positive
mentors and role models for how we could

combine entomology with philosophy so that's where we were coming from with this paper was how could we combine our Direct ethological behavioral ecological experience with the social insects with what we knew about neurogenetics and molecular biology and then tie that through respectful engagement with some of the philosophy of Consciousness literature here's what we didn't do in the paper what we didn't do is make a claim or a bet on whether ant colonies are conscious or not whether ant workers are conscious or not if people have feelings on that let's hear about them on the actm stream let's find antstream let's do something fun let's hear about it but that's actually not what we did in this paper we do a few things in this paper the first thing is that we proposed this terminology of a forward and a reverse test for Consciousness which I'm going to return to in a second with this image the second thing we did was to review the literature on Consciousness from a scientific literature and make classification schemes for different kinds of theories of Consciousness so you can see why I was interested in the breakdown of this paper's uh understanding about the fields of different scientific studies of Consciousness and then the third part of the paper we used empirical findings from entomology so observations of morphology or of behavior of individuals or of colonies from entomology and we asked whether different Frameworks for

Consciousness that we found during the
literature review might come to the
conclusion that ant colonies Andor
workers had Consciousness or awareness
and we as authors remained neutral on
the issue to the extent we could as far
as just presenting hypothetical
evaluations of colony Consciousness from
the point of view of these different
Frameworks which is something that's so
intrinsic to Consciousness so it's kind
of fun and it turns out that Frameworks
just simply wildly diverge in their
claims about ant colony Consciousness
and it's really not a surprise that it
turned out to be that way because the
same Frameworks uh diverge in their
claims pretty plainly and people diverge
in their opinions and so if the
Frameworks are going to disagree about
humans being conscious or electrons then
they're going to disagree about ants and
um it was just interesting to experience
this though and ask how we could have a
reconsideration of what we're really
doing here with the Consciousness
studies uh area because there's
thousands of speculative papers about
Consciousness in animals but are we
moving the needle and if we are in what
direction is that where we want to be
moving it what did we do in the paper so
we introduced this idea of a forward and
a reverse test for Consciousness and we
thought thought about forward in Reverse
just from the point of view of the
Observer the experimenter so that's very
related to this phenomenological

invariance idea that the PCM model is
going to return to in a little bit and
this idea that everything in
Consciousness should be relational let's
take this perspectival and this
relational approach very seriously so
what is a forward and reverse test a
forward test is when you trust your test
and it describes the mode of usage of
tool where you're measuring systems and
drawing conclusions so for for example
in the figure it says I trust my test
how heavy are these boxes and we have
this gray test it says it is an unknown
whether the box is conscious or not we
put it on there it says yep one and then
we go out in the world and we
investigate whether different boxes are
um different weights or have
Consciousness or whatever and that is
for validated calibrated instrumentation
and it's important for the normal in a
paradigm uh vocabulary the normal
science which is advancing science and a
reverse test is a different phase a
reverse test is when you don't trust
your test yet and this is like
calibration and calibration is when you
take things that you do know about like
we know that the meter is this much um
relative to um this and we know that the
wavelength of this is going to help us
calibrate this tool um but really just
when you use a scientific instrument it
has to be calibrated and if we want to
have a science of Consciousness we have
to have a calibrated instrument and so
the reverse test is when you don't trust

a test and you don't know what the
machine is going to read out but you put
on the one PB block and you say I'm
going to fine-tune this machine so that
it comes out with one when I put the one
on there and then we'll know that this
scale or this spectr photometer works
and it's kind of this back and forth
this two-stroke engine that we've seen
so much about and we've been talking
about um the back and forth between um
action and inference of course we've
been talking about that from the
framework of computationalism from the
basian perspective the basian
structuralist perspective inative ISM
and ecological psychology and just
cognition in general agency how do we do
this forward and backwards of more tool
likee based upon affordances and more
inference based upon
observations okay so it's kind of a
two-stroke engine for studying
Consciousness from a field perspective
so this is like what should the field of
Consciousness researchers be studying
now wouldn't it be great if we had a
forward test for Consciousness if we had
a scale that we could go yep we hooked
up this person's brain or this wi-fi
router or this Beetle and yeah it's a
two out of four it's a 2.9 out of you
know whatever it is yeah sounds great I
understand why people want to
mathematize and why they want to measure
Consciousness but if you meet somebody
who has a good test for Consciousness
then you know get the Tesla Tech get the

perpetual motion machine get everything you can from that person because we don't see it within the current space we're in in um we want to have this test prefrontal cortex language whatever it happens to be no caveats just a test of course it doesn't exist and so that's kind of what this issue is about and so when we don't have a forward test for Consciousness even though people might think they locally have one I hope I've sort of dispelled that with a previous slide if we don't have a forward test to operate in the action mode do we have a reverse test to help us work in the inference mode so that is um where professor soic and I wanted to think about different kinds of forward tests for Consciousness and then how could the ant colony be reversed test for Consciousness so here's the framework that we Loosely organized into for our literature framework um in terms of forward tests for Consciousness so this is like what other people might call Frameworks for Consciousness we said look all these models what they have in common is that there's some measurement some tool like intervention or um intervention into the process or measurement that could be done that would help us have a forward test so like is this system conscious does it have X does it have language does it have a prefrontal cortex okay now the reverse test for Consciousness would be like we know humans are conscious so any theory that says that

is fine with me or we know humans are not conscious so I rule out any theory that says that right whichever one you think you believe fine so we separated into a few different categories the first one was neurobiological so remember that they had three categories biofunctional cognitive neurofunctional and information theoretic what we did um with our slightly different breakdown of the uh categories in the space not a final word this is just what we had uh found out
about is we divided neurobiological models into the following categories which were uh structural models that were based upon macro or micro uh structure so macro structure would be like the cerebral cortex is this size or it has this brain region so gross anatomy macro structure the micro structure is like wow there are so many connections on the dendrites or this cell type has this shape so things that you need a microscope to see macro macro to micro so you need the microscope for the micro histology or oh the architecture of these neural networks is a feedback that means that it could be implementing something so macro micro structure of the brain neurobiology or of the brain in the body for that matter because it's not always about the brain functionalist models for um Consciousness would be like there is a function that does egocentric navigation in crawfish and crows and humans so these are conscious entities so because

there's a functional element of
something that's being done even if it's
being done in non-homologous brain
regions or two different ways by two
different species or in somebody who is
born with vision or who wasn't born with
vision it's the function that matters
and then another neurobiological
assessment that people often make in
terms of forward testing is dynamical
featur features of brain function so
they'll say well because the brain is a
complex adaptive system with winterless
competitions and leaky integration and
action policy selection and um lock of
volaric um you know uh Predator prey
type Dynamics or chaos and landslides
catastrophes power tail of noise all of
these dynamical features coming from
usually a complexity or a signal
processing perspective so those are
kinds of appeals that people make to the
brain and the
body there's also the egocentric and
relational Frameworks for thinking about
what counts as a model of Consciousness
and in this category there's a bunch
that we could go into but just to make
it kind of clear what I mean by
egocentric and relational which is
actually the first invariant principle
of the PCM so totally agree it's
critical it's um the idea of spatial
awareness and perception which is um
whether it's the lived experience or
just the acting as if of things being
closer to you being more accessible but
everything in conscious is experienced

in relation to something that's agreed upon even when people don't even agree what that relationship is or what that something is so that's kind of cool um then there are cognitive and behavioral Frameworks so these are cognitive being broadly understood as like things that computers might be able to do or computer networks or extended cognition embedded enacted encultured all this stuff that we talk about every week let's take that broadly to cognitive so here by cognitive I don't just mean what what people think of happening between you know the student's eye and them filling in the multiple choice test that's not just cognition we want to have a broad understanding of cognition multiscale message passing basian Nets Etc and the categories here are um there's multiple but examples are like emotional or affective States kind of foreshadowing um the sophisticated aec of inference which will be Act 11 but emotional states like regret um or optimism bias and cognitive biases more General so everything from uh visual Illusions to other types of illusions that are not visual but can be related to the order of things being presented these are sort of cognitive effects that potentially like a neural network might fall victim to and so what we've experienced is that there's this runaround and people will say well this bird has this brain region so it's a macr structural claim so it's conscious and then somebody brings up a a shortcoming of the idea of macr structuralism like well does it really mean that it's aware just cuz it has this brain region what about sleeping people or what about you know Daniel dennet or something like that and they go well right it's actually the brain region and the micro structure okay so it's micr structural great well then would anything with that micro structure have that kind of a characteristic um like if you made a model in a computer we did have awareness well no it has to be evolved or it has to have language or it has to be implementing something ecologically great so we want to chase that rabbit down because we're serious about answers here and so we want to chase it down someone says oh well when I said language I didn't just mean a chatbot I meant they they they feel emotion you say well but what if the chatbot says it feels emotion too what are you going to do and then all of a sudden that whole feeling emotion um communication of a feeling emotion all of a sudden they have a new explanation so let's keep on chasing down these rabbit holes and uh one other category that is closest to what is described in this paper is the mathematical framework so we took a step beyond the authors in the um 2018 paper the projective Consciousness model where they said there's the information Theory category of Consciousness models we actually separated a few of these models of mathematical Frameworks so we had

decision- making and long-term planning
um as well as information the theoretic
and GE geometrical so there have been
some uh maybe it was published similar
or slightly after this paper but there
have been some uh geometric Theory Of
Consciousness info uh Theory and then
also a lot of times the information
Theory approaches deal with information
geometry so which one is it I agree that
it's enough to call it mathematical and
just say includes topology information
Theory thermodynamics geometry just a
lot of math stuff um another area of
mathematical Frameworks for
Consciousness could be like cybernetics
which is why it was one of the keywords
for this paper because a mathematical
framework for Consciousness might
involve decision-making or agency or
ability to plan at a certain level of
coherence and in fact in the 2018
interview that was with Carl friston
myself and our pass away colleague
Martin forer in the last question of
that interview I asked him whether he
thought the ant colony was conscious
because I was working on this paper with
Eric and I was writing the interview
with Carl and then I just thought wow we
could figure out what Carl thinks about
ant colony's being conscious and if
you're interested in either fris
andology or in ant colony Consciousness
then definitely review Carl's answer on
that but just to kind of close this
slide off um we're committed to
long-term Clarity on these issues about

the study of the science of
Consciousness so we really welcome
collaboration or Improvement on these
questions let's work with it let's make
an ontology for Consciousness or how are
we going to get organized on this issue
because as I said there's just too much
on the table to let this one slide and
we have to enter this space and do it
right because otherwise just it goes off
the rails
okay let's get to neurophenomenology
which is one of the keywords of the
paper oh yeah the paper and on the
bottom here I have a couple just fun
images the left side is the book GB by
Douglas hoffstead 1979 but reads as
crisply as ever and uh really an
insightful book and then on the other
side uh of uh is the tree of knowledge
this yellow cover by verela and matara
and that's a book about autop poesis and
a few other topics that are really cool
related to complexity and just very
deeply related to the ecological an
activist semantic biobehavioral lot of
great stuff also potentially the um work
of frof cpra is one thing to look into
here but then some images because it's
not all about language so here are some
images by mcre and by mcer um that have
to do with sort of what is this self
let's look at how the authors Define
neurophenomenology they Define it as one
observation and description of the
phenomenological invariance at the
appropriate level of abstraction so
maybe the appropriate level of

abstraction um for the experience of
drinking water is not uh you know a
throat cell maybe it's uh not a whole
nation maybe it's uh a person the
classification to classification of the
behavioral cognitive and biofunctional
aspects of Consciousness so some drugs
like you know you take acetamin and it
doesn't make you feel that different
there's other drugs that are
psychoactive or that have long-term
changes to your brain whether you
experience it in the short term or not
so how are we going to get a behavioral
cognitive molecular genomic
understanding of that
mechanism three identification of the
neuroanatomical and neurophysiological
correlates of Consciousness so I see
neuro Anatomy as the structural whether
it's F uh fun structural at the macro or
the micro neuro Anatomy is what sovic
and I called structural then
neurophysiological is um referring to
the Dynamics and the processes molecular
informational whatever they may be that
are in the brain so those are the
functionalist models to me of the brain
four development of mathematical and
computational models of Consciousness
which I would consider to be in the
mathematical category that a Faithfully
integrate all the invariants described
in one which we'll get to soon B provide
unifying explanations of inestable
hypotheses about the biofunctional
aspects of Consciousness described in
two all on board and C can be physically

implemented in the hardware identified
in three as well as perhaps in
non-biological Hardware so I don't
subscribe to computationalism uh I don't
believe in a hardware software
distinction except as a useful metaphor
in biology I think the book wetwear is
pretty interesting on this topic and the
reason why I intervened right after the
mathematical and computational models is
because I think it's actually a
different domain I think there could be
a mathematical model that's extremely
agnostic to the biology or our
biological model could just be like it
is what it is and it doesn't matter what
the m or what anyone thinks the math is
it is what it is because of evolution or
um because it passed through some sort
of historical filter and any equation
you put on it is just like a linear
regression it just it doesn't matter so
those to me they they can kind of fly
together and that would be great or not
it could be something totally off base
which is why I think again it's
important that even though we have the
citations and the ways that we can sort
citations into keyword clusters let's
not think that we've mapped out the
space of the possible in consciousness
because there's people in other cultures
who don't feel the same way as
scientists first person perspective is I
think the last keyword and they write
about that in the paper they say ideally
this method of the PCM would give us a
theory of Consciousness that

intelligibly links the phenomenology of
a firstperson conscious experience with
its realization in the brain or other
substrate via mathematical and
computational model simultaneously
accounting for its biofunctional
properties so the B functional
properties is kind of where you got the
um neuro part the mathematical
computational model they're saying
wouldn't it be good for purposes of
describing it predicting it controlling
it designing it um explaining variants
reducing our uncertainty why not that's
the mathematical computational side and
then pull it all the way back
phenomenology so computational neurop
phenomenology we're going to be doing
mathematical computational neurop
phenomenology in this paper and we're
going to be using geomet tools like
projection mapping to think about how
cybernetics and active inference are
related to Classic work in neuro
phenomenology okay one more um key word
perspectival imagination so they quote
on this one they write this perspectival
space of the field of Consciousness FOC
is organized around an elusive implicit
origin or zero point and includes a
horizon with Vanishing points at
Infinity that Mark its limit so right
there there's the two of the violations
of ukan geometry point of privilege
that's the point of view and then the
vanishing Horizon points at Infinity the
FOC embeds vectorized directions that
frame the orientation of attention and

action along three axes vertical
horizontal and sagittal so like you know
this one this one and then that one
these are given in perspective with
respect to these same points at Infinity
so pretty interesting stuff they
continue and they write since we are
dealing with a 3D perspec hial space
with an elusive origin point of
privilege point of view and Vanishing
points at Infinity constituting a
horizontal plane a horizon plane that
happens to be in the horizontal axis
doesn't have to be if you're on your
side it's
not the PCM postulates that the field of
Consciousness has the structure of a 3D
projective space in the sense of
projective geometry nice thanks for
giving it as a keyword too it helped me
learn about it the FOC can thus partly
be understood in terms of a 3G
projective frame
let's look at a few just kind of cool
examples related to perspectival
imagination so this is an example of a
non- ukian geometry it's an eer uh
artwork and it's like these bats that
kind of fractal off and sometimes this
could be related to hyperbolic geometry
or other geometries but the point is
like each bat might be flying off and so
that kind of maps onto our experience of
a bird or a bat flying away from us and
slowing down perhaps um but how is that
going to get mapped onto another space
that may or may not have three
dimensions spatially again Bucky Fuller

might claim there are four spatial Dimensions so let's not assume there are three spatial dimensions and that the fourth one is time that being said it makes sense to talk about XYZ coordinates from a cartisian perspective another example about this perspectival imagination is vanishing point so everyone who's done a fun drawing course or just uh looked down a city road or looked out of a car that was driving through Fields everything kind of vanishes and so you can see the rows of a field and you know that they're all in parallel but then you can see really deeply down one other ones like kind of move towards it and another way of seeing it in a cityscape is like you can conceptually know that the street is straight but if you look at it it looks like a triangle that's disappearing from you with a point at the top so what is it that allows us to be like oh yeah buildings have 90° angles and they're going upright even though all the actual angles are kind of whack

so that's this perspectival imagination and that's what makes an artist who can make a lifelike drawing especially like an architecture or something look lifelike because they capture these extremely subtle features about vanishing point and perspective and all these different things so one drawing that I love on this is um this

one and it's like for many people it's said okay do a self-portrait of what you see when you um are on the couch and one approach might be like I just you know draw like okay I see this you know here's this piece of art here's my bookshelf here's you know my bike then the next layer back like oh I'm going to include my feet in the drawing like the kind of you know I'm on the beach here's my legs type thing like uh taking a picture from right in front of you with a phone something that we actually have an affordance to do now but other people may not have seen in such a direct way um which is why they didn't take selfies for example oh hey here I am this drawing takes it back another layer now this is actually where are you really seeing from not what is canceled out with a binocular vision but what happens when you look with one eye really really far to the side well you see this curving and it's just like a little bit beautifully accentuated but of course we're seeing that as a two-dimensional projection so that's just so cool and uh styles that play with our perception like optical illusions or cubism or abstract art they often incorporate some invariant features of perspectival imagination but then violate expectations on other ones so like um half of the person's body is as if they were lit from this side and half is if is as if they were lit from a different side that was a classic technique of the cubists and so very rich space very open for honest collaboration with artists and philosophers and practitioners so we hope that this is kind of an opening for those who might have thought active inference okay it's an engineering framework we're talking about optimization we're talking about sociocultural things but kind of from an abstract point of view how are we going to take this into our experience of Consciousness and of perspective so I'm going to return to this slide again and again which is that it's a question what does free energy principle and active inference say about the relationship between the world and agents so that's this feedback between the world and the agents and the nested agents and here we're adding on this level of the agent is having conscious awareness it's thinking I am a strange action perception Loop or at least it's acting like it says it or it's thinking that it's saying it and just as we've talked about before with this slide we want to be building on the fields of action of inference of physics of experience we want want to be tying these threads together while also being aware that just being convinced will never be enough so that's why it's such a fun area why it's so great to do research in this area and to learn about it and also to welcome collaboration and to welcome people sharing their perspective because I really believe that no matter how many papers you've read about Consciousness it does not make that opinion any more valid there's other topics where

potentially you couldn't say that as
flippantly but in the area of
Consciousness it can't be a counting
game it's just not it's something that
is so essential to our experience and so
it's such a deep area a perfect area for
transdisciplinary
synthesis cool let's get to the paper
because it's also all about the
specifics we want to be really specific
about what the authors say and do so
that we can build off of it with other
specific work so that we can actually um
use their perspectives as catalysts for
our own development not push ideas out
of our head and replace it with you know
the frisen update but rather be
catalyzed and spurred to new culturally
informed creative states by reading and
by discussing so that's why really
especially for 9.1 and point2 for the
last home stretch of 2020 and for 2021
just if you're listening to it we want
to know your experience and we want to
hear your questions in the YouTube chat
or have you on live however it has to be
this paper is called the projective
Consciousness model and phenomenal
selfhood it was published in December of
2018 in the uh Frontiers and psychology
journal in the division of theoretical
philosophical psychology and the authors
are Will aord benoquin friston and redr
so that's kind of giving you some
information on when this paper was
published and on what areas the authors
want to place their work within let's
take some quotes from the paper and look

at the main aims and claims of the work they write can we describe this integrated multimodal and centered spatial structure of conscious experience in a rigorous way great question I love the setup and they WR and can we give a good account of why it should be thus organized here we answer both questions in the affirmative great questions I love the setup and it's something we can evaluate whether that's a hearty affirmative or a half-hearted or a overly confident or underly confident or okay it's affirmative but then what do we do these are all great questions to have and I just love that the authors can be really clear about this great research and also a great suggestion by a Community member to um read this
paper overall the PCM delivers an account of the phenomenologically available generic structure of Consciousness and shows how Consciousness allows organisms to integrate multimodal sensory information memory and emotion in order to control Behavior enhance resilience optimize preference satisfaction and minimize predictive error in an efficient manner sounds chill if we could we should great the way that I would rephrase it without um any of the uh jargon of actm is how can we apply active inference to the study of Consciousness in all of its unified yet coherent mysterious Dynamic and profound facets so there for the purposes of scientific writing

highlighted a lot of the vocabulary
keywords and research Fields related to
control theory information signaling
geometry math computation those are some
heavy-hitting ideas but let's also not
lose sight that some of the other
heavy-hitting ideas include the mystery
of Consciousness and why we're the one
that we are and whether cells have
Consciousness and whether ant colonies
do and whether colonies do but not
nestmates and nestmates but not colonies
or ecosystems but not colonies just
let's be open-minded so let's answer
questions in the affirmative and be
open-minded let's have yes and and let's
accomplish yes and with everyone's
participation the abstract begins with
saying we summarize our recently
introduced projective Consciousness
model and relate it to outstanding
conceptual issues in the theory of
Consciousness the PCM combines a
projective geometrical model of the
perspectival phenomenological structure
of the field of Consciousness with the
variational free energy minimization
model of active inference yielding an
account of the cybernetic function of
Consciousness viz the modulation of The
Field's cognitive and affective Dynamics
for the effective control of embodied
agents long sentence double double
points for um using uh affect and eect
uh long sentences but slow it down
unpack it I think that we've gone
through the keywords well enough at
least a first pass to show you why it

would be great to connect across all these areas abstract part two the geometrical and active inference components are linked via the concept of projective transformation which is crucial to understanding how conscious organisms integrate perception emotion memory reasoning and perspectival Imagination in order to control Behavior enhance resilience and optimize preference satisfaction so sounds cool the PCM makes substantiative empirical predictions we'll see about that and fits well into a neuroc computationalist framework of course it does it also helps us to account for aspects of subjective character that are sometimes ignored or conflated pre-reflective self-consciousness the firstperson point of view the sense of mindness or ownership and social self-consciousness and close the abstract with saying we argue that the PCM though still in development offers us the most complete Theory today of what Thomas metzinger has called phenomenal selfhood now I don't know meter's work incredibly well or what this topic Bridges out to kind of an interesting way to end the abstract but so it goes um that's what we might be getting to how are the authors going to get from all of these keywords like cybernetics and geometry to getting to potentially the most developed framework to date in their opinion of what is called phenomenal selfhood so phenomenology experience selfhood having

a self
well they start in section one with the introduction and go pretty directly to section two methodology I think this is great because it really gets into the details quickly you could philosophize for a thousand books people have so what's the methodology how can I identify what you're actually doing in this paper as a researcher um without hearing 30 pages about your understanding of what DART did or didn't say three they're going to talk about phenomenological invariance and the functional features of Consciousness which are also we're going to go into them later and they talk about the phenomenal logical invariance what they are five of them we'll go through them and then talk about how they're related to projective geometry in the PCM so basically they're going to start with their invariance and if you have a qual with an invariant you want a sixth one you want to collapse to four great let's write another paper but they're going to start with these invariant features and then they're going to introduce that plus projective geometry to say projective geometry is the bridge we're going to take to get these five invariant features we see figure one which has an account in the PCM model of psychological phenomena and figure two which shows projective geometry in relationship to the Necker Cube which is like kind of a cool illusion that I'm looking forward to showing you then they

go to the five functional features of Consciousness which are also what we're going to address and they talk about the projective Consciousness model which is remember the phen phenomenological invariance plus projective geometry equals part of the PCM now we're going to talk about the PCM and the functional features of Consciousness the functional cybernetic functions that they talked about earlier and fuse that in the capacity of using active inference and free energy minimization to get some of these functional Dynamics then they'll show figure three a simplified 2D um projective Consciousness model based simulation of a pit Challenge and then figure four give a overall sketch of the projective Consciousness model
in section four they talk about the hard problem of Consciousness representationalism and phenomenal selfhood the hard problem and computational functionalism is um variously phrased by uh professor chers and others as basically being about what the substrate of Consciousness is why is it that when you hit your hand it hurts but when you hit some brain regions uh it doesn't hurt and when you hit some other brain regions you just pass out you come back or you die or something in between how is that happening in the brain or is it an antenna is it a generator these are the
questions then they go from the computational functionalism to talking about the PCM and representationalism

now all these terms computation function
and representation are pretty loaded
we've gone into them on a couple of
other discussions but be open-minded as
to how they could recombine in the
context of PCM and then they're going to
take those kind of more quantitative
terms computation function
representation representation is a
little bit of a bridge and they're going
to bring that into the projective
Consciousness models phenomenological
components so that's going to be about
subjective character subjective
character being what it isness the what
is it about to be
about and also the the phenomenal
selfhood of the philosopher that they
concluded with discussing in their
abstract and then they conclude so looks
like a great road map let's in this
introductory video in the um time that
I've left here let's go through the
functional features and the invariances
of Consciousness because we're not going
to have time to go through them in 9.1
or 9.2 with all of you contributing your
perspectives and it's good to get them
out here because these are kind of like
the big five for the functions and the
big five for the invariances and if you
have a qualum with the model you should
first ask is there a function or is
there an invariant that I think is
extraneous um is there one that I think
is missing is there one that I is just
off base so if you have a question about
one of these five things you think this

doesn't go far enough or this goes too far or but some conscious systems don't have that great that's the question you want to ask so there's five in each category if you think there should be more or less or a different one that's one entry point now if you agree with all them and you say yep those are the perfect five in each category then you can evaluate given what projective geometry is and what the PCM is do they May address the functional and the um uh experiential invariance of Consciousness so that's how we're going to read this paper and that's where I hope there's a lot of footholds for people who maybe have thought about Consciousness but this is a new way to hear about it for them it would be great to have their comments what are the five functional features of Consciousness according to the

authors and the way that they sort of justify this is by saying um we claim that the overall function of Consciousness is to address a general cybernetic problem or problem of control control theory Consciousness enables a situated organism with multiple sensory channels to navigate its environment and satisfy its biological and derived needs efficiently this entails minimizing predictively erroneous representations of the world and maximizing preference satisfaction a good model of Consciousness must be able to explain how this is accomplished the PCM thus emphasizes the following interrelated

functional features of Consciousness
here's why I pause before putting it up
but after just in it because even things
that do control theory might not have
Consciousness full stop that's the whole
thing that I was getting into with the
ant colony test is okay so then if I
have a robot and it has something that
allows it to do multimodal integration
across sensors with a generative model
with a deep with a counterfactual
whatever it is does it have
Consciousness and then that's where the
person goes oh well it needs a brain and
so then they go down the rabbit hole a
different way so the function is
important but again again it's not the
whole story especially yet here are what
they characterize as the five functional
cybernetic features of Consciousness
that we'd want to at least explain so
maybe just passing these five is
necessary but it's not sufficient but
it's important one Global optimization
and resilience Consciousness helps the
organism reliably find the globally
Optimal Solutions available to it given
competing preferences knowledge based
capacities and context making the
organism more resilient this is a
mechanism of global or subjective here
in the of subjective basian optimization
which does not always obil objective
optimality though it must approximate
the ladder reliably enough uh or you die
if you don't objectively enough figure
out how to walk down a staircase you're
not going to walk down many staircases

now a few things on this first many
people think Consciousness is epif
phenomenal so whatever is happening in
our awareness is not going back to guide
so this plays a little fast and loose
because it's saying well Consciousness
helps your organism reliably find it
sounds like Consciousness is is coming
back and this is the whole cartisian
dualism Paradigm is like I know that
Carl friston and others have many more
nuanced Works including I think
therefore I am related works so I'm not
going to go into it but it's like if
Consciousness is something that reaches
back into the physical how is that or if
Consciousness is purely a descriptive
epip phenomenal epip phenomena above the
phenomena of mechanistically or um
materially defined components like
neurons message passing whatever it has
to be ants no no matter how emergent
their properties are in the weak
emergent sense do they ever amount to
strong emergence and actually cause the
system even the um downward causation
and causal entropy and the
renormalization group that still is a
very philosophically agnostic with
respect to whether it's causal so
another thing to say here is that Global
optimization given the preferences
knowledge based capacities affordances
and context so so it's not Global
optimization so it's local optimization
or its Global exploration of a
conditional space conditional upon the
things that we want it to be conditional

on like how far you can reach but then I
don't know about you all but my
Consciousness doesn't always help me get
to the globally optimal points unless
that Global Optimum is defined as this
not objectively best space of like well
you you just didn't go the right way but
it seemed optimal given what you knew
it's like okay and then okay and it has
to be reliable enough so I'm on board
two Global availability the contents of
Consciousness must be generally globally
available in the sense of poised to be
treated of by our effective cognitive
and perceptual systems as the occasion
requires so for example you know here's
idea 1 idea 2 memory 4 the color you
know the number all these things are
just Apparently one layer away from
accessing the workspace this is drawing
heavily on the global workspace Theory
attention schema model of some other
researchers I would just point out
Global doesn't make sense to me the
brain isn't the globe cognition isn't
the globe availability it's like yes but
then again you define available as like
I knew it yesterday but I don't know it
now so it's not available um or what if
it's part of my extended cognitive
process and it just takes me five
minutes to find it so okay relevant but
globally available um and also our
awareness is hardly the tip of the
iceberg um or maybe it's
a whole Iceberg and a half so I wouldn't
even know what it would mean to be
globally

available motivation of action and
modulation of attention attention schema
Theory Consciousness is generally
implicated in the motivation of action
based upon affective Dynamics and memory
this drives the modulation of attention
though Consciousness is not to be
narrowly identified with attention so
that's kind of attention schema Theory
Breaking Upwards out of the just um like
uh lstm like a neural network that has
an intentional parameter we want to go
into the experiential Dimension not just
the oh well it's a laser that spends a
lot of its time here or um you know it's
a camera that autofocuses on this level
we want to go beyond that and say um
whether or not it is getting input like
a camera in some of its sensory organs
the experience is somehow generated but
I'd also point to this motivation of
action and modulation of attention it's
very very important these are phrased as
if Consciousness functionally functions
of Consciousness right motivates action
and modulates attention or it could be
epiphenomenal Consciousness that is and
so it could be the experience of
motivation of action and the experience
of modulated attention so in that
framing we made Consciousness
epiphenomenal and we basically said yeah
attention is something that happens with
all the neurons and the firing patterns
and whatnot you know the drugs
everything that happens and then it gets
to some synergetic compromise where it's
like looking at this thing at this time

and then you're just experiencing it
it's not simply a cartisian theater
model it's still a bottom up top down um
situation but there's no simple outs
here it's neither that Consciousness is
reaching back and motivating action
neither is it that it's modulating
attention through what dials you know
show them to me oh well that's what
meditation is yeah but then all of a
sudden you're getting into this area
where I think a lot of stuff that's not
related is pretty
important and then the fourth one oh
whoops I said big five well let's just
think of it as the big five the big
fifth function is going to be the secret
function but the fourth function is
simulation enhancement maybe that get us
too Consciousness facilitates the use of
simulation in the broad sense of
imagination not the narrow sense like
simulation Theory um but maybe that's a
direction to go in the service of
solving cognitive perceptual and affec
of problems and more generally in the
anticipation and programming of actions
and the response to their outcomes so
simulation enhancement could be for
example um I wake up and it's very dark
in my room but I have a generative model
um or an accessible model not um being
too formal here of my room so even with
my eyes closed or with very dim light I
can enhance my simulation and think oh
wait you know if I open the door this
way it's going to wake up my roommate or
if I do this at this time then down the

road this is going to be a different
sensory input that I don't want to
experience so that kind of Rich U
mechanistic understanding of the world
is what humans excel at and it's what
the current machines that we deal with
do not deal with well um you can ask it
a question like could you know this
building wear ice skates and it might be
able to parse whether um those terms
mean different things and whether
they're keywords used by the same people
or not on Twitter but will it be able to
understand and address that question and
that's kind of where the GB book really
is fun to think about so again four
functional features for Consciousness
should there be a fifth one are one of
these extraneous do you have a different
function for consciousness what is your
understanding is consciousness have
function or is consciousness something
else how would it affect things if it
had a function how would we measure the
function what is dysfunction all these
questions any questions that people want
to have maybe be cool for people who
watch it like to post questions they
have and we can have a norm of just
asking the questions that we have out
loud
because let's ask the questions we have
I don't know any other way to say it
right
now the second thing that will we'll
talk about is the um invariance of
Consciousness the phenomenological
invariance and here I believe there are

five so these are the things that are really what their model is going to come down to with respect to phenomenology and philosophy so this is what they're going to focus on this is what the mathematical model needs to aim for this is what their formalisms should explain predict integrate Encompass um so this is like the big core philosophy idea that we want to do all of the math and all of the computational stuff to get at with some type of bridge the bridge is projective geometry perhaps so what does it mean to be a phenomenological invariance and they write the following are the deeply interconnected invariant that we think characterize Consciousness we take them as postulates for mathematical treatment citation that you could follow up on and construct the PCM on their basis so it's kind of like axioms like they're taking these as like roots statements that kind of don't need to be justified they're kind of like observations but of course this is something that we all want to know is could there be another one could one of these be redundant what if you have four out of five do you get 80% on Consciousness or what if we figure out or Imagine A system that has uh three out of five is it you know what does that mean for that system what are these phenomenological invariants one relational phenomenal intentionality so break it down by the words all Consciousness involves the appearance of a world of objects properties Etc in

various qualitative or representational
worlds representational ways to an
organism so again they're not thinking
too much about worms and fruit flies and
ants and ant colonies maybe they are I'd
love to hear it from an author or from a
colleague if they are but um these are
the debates that we're having which is
like right if You observe the worm
moving away from the gradient as if it
had a representational relational
phenomenal intentionality
is that evidence enough and um that's
the question two situated
three-dimensional spatiality of course
also there's time sometimes called the
fourth dimension the space of the
presented world like objects or Etc is
three-dimensional again big asterisk and
perspectival unfolding in an oriented
manner between a point of view and a
horizon at Infinity we all parallel
lines
converge the origin of this point of
view is elusive though normally it seems
to be located in the head the space is
normally organized around the lived body
so this is like bringing us to
embodiment to in activism to ecological
psychology to Intercultural and
developmental psychology to um Ken
Wilbur's integral Theory also connecting
us to lived experience and to spatiality
and to the ways that the spaces that we
live in influence who and how we are so
all of the awesome ACM stream
participants who have taught us so much
about embedded systems and embodiment

we'd really love to hear what is work in
this area that impinges on one of these
invariants multimodal synchronic
integration synchronic means at the same
time Consciousness involves the
synthesis or integration into a unified
hole of a multiplicity of qualitative
and representational components from
sensory modalities memory and cognition
so dionic means at a different time
synchronic at the same time so memory is
dionic but it's of sensory modalities so
it could be a hearing event from six
months ago um maybe you know and there's
the whole question of imagination
obviously big question but just to a
first pass um events from the past that
influence us in the future are dionic or
another way to think about it is they
modify the system or they put it onto a
different trajectory that's different
than if that differen making cause to
use the waters terminology hadn't
occurred Consciousness seems to be both
dionic and synchronic like when we're
listening to music and seeing flashing
lights there's like synchronization in
the same moment that's also
multi-sensory but there's some really
interesting work about the time
durations of different kinds of
experience so clearly something is
happening there with Sight and Sound and
just people who are born with or without
hearing and born with without sight all
these different people can navigate in
the same world so that says something
about sight you know it's not just us

watching a projector if there's people
who can navigate without sight so that's
kind of what they want to bring it back
to four temporal integration which I
just hinted at before but this is a
little deeper level Consciousness
involves at least the retention of
immediately past experiences and
pretention of immediately future
experiences as integral elements of its
specious present the uh everpresent now
as it were and a foundation for more
expansive forms of temporal ation such
as Distant Memories and long-term
planning distant past distant future now
just to give one kind of funny note when
I was reading this I thought what is
protention what does it mean in this
context so according to Google
protention is a philosophy word that
means anticipation of a future event so
I dug a little deeper into the Mariam
Webster definition of prot tent P R O T
N T Pro tent like a professionally setup
tent it's an archaic transitive verb
meaning to stretch forth or to extend
now nowadays one might hear about
portending future events not penting
them um and the words actually have
different meanings so to portend p o r t
n d portend is to um convey something
through like a symbol or an omen like
the the owl portended difficult times or
something like that but um port tenting
is something that the agent does in
response to the symbol so um upon seeing
the owl you know the farmer decide
pented that this was going to occur and

so I would just kind of bring up this
idea that maybe they're so similar
sounding and they're so kind of close in
their word usage that the English
language just couldn't tolerate um that
much ambiguity because penting and
portending they sound like 99% identical
and so then maybe portending took
popularity over prent because it removes
agency from the actor and so it's one
way to remove action from language and
so now proten or
penting it's now a philosophy word it's
archaic it's an old word because we
can't have action in speech right no but
non-ironically it is really interesting
to think about how these words change in
meaning and the more action we can embed
in our speech patterns and our
interaction patterns the
better subjective character um so
subjective characters defined by the
authors as involving a pre-reflective
non-conceptual awareness of oneself and
their individuality now one could be
totally off base so they could be wrong
is that part of Consciousness to be
wrong about your individuality or is it
just that you're having an individuality
out uh hard to say but the authors have
it as an invariant they write also that
two to five are in fact implicated in
one so relational phenomenal
intentionality is really the most
important key invariant so that's the
relational thinking and that's why I
feel like the active inference and free
energy principle communities are very

adjacent to complexity thinking to
systems thinking relational thinking
design thinking as well as um team
collaboration Frameworks because so many
of these social work and team relation
Frameworks as well as other types of
thinking including complexity they all
deal with this relational Insight so
there's so much to learn and to do here
and to understand about what are the
implications of relational thinking on
Consciousness on our own experience if
not on actually making claims about
Consciousness cool let's just kind of
close out by going through and looking
briefly at a few of the figures and just
what they're about take a 30,000 foot
view so that when you're reading through
the paper you kind of know where you're
coming from so figure one has um a or
and c and and um a is a projective
solution to a VR VR mediated
disassociation in phenomenal selfhood so
this is kind of like um from the
experience of uh the ukian view like
we're looking at Minecraft here and then
the person their head is in this like
Block in the lived space in the right
side of figure a and um but they're
Imagining the space as if it were on the
left side of a so I can imagine my
office right now now in terms of the
right angles again even when I look out
I don't see right angles if I were to
combine like this and I look back down
and I look at the angle that the room
was just making it's a not 90° angle but
it looks normal within my generative and

usually verified experience that it is
90° because why projective geometry so
what do we have to do well it turns out
to add in this element of the visual
field in the projective geometry in B
we're going to show this like sort of
they call it an anti-space beyond the
plane I'm not even going to go into what
that may or may not be because I don't
want to um go down too deep but it's
kind of like there's some mapping
reflected in B and C what they call
God's eye Vantage Point there's certain
mappings from B and C just like there's
a mapping between the understanding that
I'm in a ukian office and then my lived
experience where apparently wherever my
eyes are or something like that that
right a little bit behind there a little
in front of there
um how are we going to square the circle
almost literally with the generative
model of the ukian office with my lived
experience of the wacky angles and
what's interesting here is the visual
focus and so where are all these
different centers is this kind of
related to like what if we feel like our
emotional Center is somewhere else in
our body or what if we um have the
experience of not just a simple out of
body like floating above your body
looking down but more expansive
experiences so would love to go into
that phenomenology with someone
potentially from Alias or any other
colleague um and so there's certain
mappings just like ukian to non- ukian

office there's this like God's eye
mapping you can evaluate for yourself
whether you think that's God's eye
figure two is a projective solution to
the Necker Cube and so here's what's
happening in this image on the left side
is like two versions of the so-called
necro Cube illusion so it's a cube
visual illusion and it turns out it's
illusion um first off just looking at
these two for one or for both just ask
yourself do I see them as um how should
I even say this the biggest side that's
close to you is it on the top left of
each square or is it on the bottom
right and you can flip some times some
people different people in different
ways can flip in their mind seeing it as
if the biggest side of the Box were
coming down to the right or whether the
biggest part of the box is kind of going
up into the left okay now here is why
they are suggesting that this um visual
illusion exists because there's no
perspectival cues we're looking into a
projection on a flat
plane and so there's sort of like two
Optimal Solutions to the perspectival
question and so because there's two such
similar um evidence competing models
free energy minimizing models it's B
it's not they they it's undecidable free
energy based inference which is to say
that in our optimization perception we
continue just oscillating it's like you
know it's the old woman it's the young
woman it's the old woman it's the young
woman okay that's a visual illusion but

this is one that has to do with
projective geometry
where it's like it's coming towards me
like but I'm getting sensory evidence
it's going that way in a different
simulated frame and then you perceive it
that way and now it's top down so it's
like this um back and forth resonance
and there's a lot of interesting
research on this what breaks the
Symmetry quite literally and the answer
is lived experience from the point of
view simulation so here if I'm closer to
one side of the square I don't have any
little squares to pick up but it
resolves it not absolutely you still
know that square has a different side or
the cube has a different side and you
can even imagine what it might look like
like oh I'm seeing the six is on the die
it's closest to me if I were on the
other side I would see a one and it
would look bigger and I wouldn't be able
to see the six but where I'm at I just
see a big rectangle six or something
like that and so that kind of captures a
lot of these aspects of the geometry of
Consciousness where you can simulate
being in different places but you're
always in a certain SP spot visualizing
it what is that spot that's the
privilege point of view that's the point
of reference what about it is special
well it's like a non-arbitrary point in
a projective geometry what kind of
projective geometry well something that
takes our allegedly ukian world and Maps
it into this like vanishing point

experience I think I got some of the
main uh threads to link there but that's
kind of what they're suggesting and free
energy is going to be the optimization
and action and inference Paradigm within
a simulation broadly construed
understanding about how symmetry is
broken in terms of experience so big
stuff interesting topics you know to be
wanting to address this is big by the
authors figure three is a uh PCM based
simulation of a pit Challenge and on the
very bottom you'll see this is like from
another paper so if you're curious about
this simulation definitely go check out
the other paper that they reference but
what we can look at here is um starting
on the left side so there's like a bunch
of barbell shapes and basically the
barbell uh has to do with it's like a
map they can walk around in so they can
wander around but they kind of have to
make it across this pit to get to like a
rewarding place um so there's an
anhedonic
in a safe space a challenge room and a
goal room I believe is what they call
them and then they're going to be
thinking about that that moving circle
like let's see if I can do this there we
go this moving circle is like the agent
moving through the map here's its
expectations and where its prior beliefs
are and then here is it resolving its
uncertainty in a it's kind of like looks
like a radar dial but that white spot in
the middle is like the the nonarbitrary
origin that's like the Consciousness

Zone and so from the POV point of view
of this organism in its FOC its field of
Consciousness which is like some
geometry mapping from the um experienced
space into its generative model it's
navigating that mapping somehow through
Consciousness and then that's also able
to bear on the different between like
having a weaker imagination and a
stronger imagination they have this
active inference Engine with um multi
sensory integration and this looks like
something definitely to go into for the
9.1 and for our own future uh research
but basically we can run model scenarios
we can say yeah I guess if it it's a
funny thing to say you know I can
imagine that if you were to do this
experiment you might find inform ative
results suggesting that enabling an
agent to have imagination in an action
oriented geometry might enable it to do
better than a model that doesn't
therefore spending most of its time
learning about relationships that may or
may not be relevant in the geometrical
frame that sensory data actually occur
in again long sentences abound but I
think that um there's a lot there in
figure three and also a lot to learn
about how they're framing free
energy all right last figure figure four
figure four is an overall sketch of the
PCM the projective Consciousness model
and I think uh there's a few different
dimensions here so one is the Observer
at the center with their lived space uh
Stephen this really reminds me of the

Parry personal space so I hope that
you'll have a lot to teach us in this
domain and within this domain this field
of Consciousness field of affordances
let's think about it from many different
perspectives just by saying one thing
I'm not always meaning to limit I'm just
saying that this is one way we can think
about it let's
imagine that there are perceived and
imagined events now observations are
just models of observations so maybe
there's less of a distinction there than
might be seen but in this geometry
Observer Centric with all of this
projective geometry mapping happening
can we think about aligning your vector
through time down a gradient that goes
from red to Green so it's funny because
it's a gradient that colorblind people
of certain types may not be able to
perceive which just so perfectly entails
everything we're talking about here
which is that um action in some like
absolutist sense is moving downhill
thermodynamically there's very few
people who would say that action moves
uphill thermodynamically bring them on
the Stream but it has to go uphill
conceptually or symbolically in certain
cases if not just simply logistically in
terms of going uphill to go downhill
later and so here we see a gradient
reflecting the free energy going um from
a culturally based green is okay and low
surprise and then red is high surprise
high alert high attention where have we
seen that before um taking that kind of

a culturally imbued color scale and then aligning a Time Vector as a dimensional projection onto that figure as a way to say that actually through space and time in the context of the projected geometry and affordance capacities of the organism it is still downhill in a sense so I don't know where this um necessarily fits in with uh all the mathematical formalisms here looks pretty nice to talk about like the action perception prediction imagination Loop this is another loop we might want to learn about we've talked a lot about loops related to like agent and environment but now we're almost thinking like the agent within the agent is going from action perception prediction imagination so that's something happening inside of us potentially and then um again on the right side is looking at time series that's the directions we want to take all free energy research is like through time through space any other dimensions we come across we'll go through those too but that's definitely the cool Dimension to think about is deep causal inference with the sound and what's visualized um in the figure so other people may see very different things in this figure and we'd welcome hearing about it let's close with some just um final thoughts and questions to end on which maybe we'll start ending with questions more they write in their paper computationalist a poori that means after the fact identity Theory allows that apparently contingent processing constraints may be essential to real Consciousness as long as the computational model captures the phenomenological invariance meets the functional constraints and survives the tribunal of prediction and experiment we should happily accept that some apparently contingent and only empirically knowable constraints pertaining to the physical realization of Consciousness are actually essential to it so that is basically like saying look if we do the Phil if we can explain and predict the invariance the five invariant and we can capture these functional attributes there's four of them but again both of those sets the invariances and the functional attributes anyone is game to add remove or change it just needs to be justified and understood so they're saying look if it does both of those things which is why I spent so much time on it in this talk was because that's their claim it's not just broadly we could do active inference and think about Consciousness it's saying if we make these invariances part of our model and we at least get all the functional attributes that we want then any model that is surviving unique explanation and predictions the tribunal the uh the tra as it were if it does all of that then it is basically going to come along with some wacky baggage like a parameter we don't understand or the God's eyee view and then they're just saying you're just

going to have to accept that as
contingent so fun thought by the authors
just a really well written and conveyed
paper and I hope that it broaches onto
many areas that we've talked about in
other actm streams here's my closing set
of questions one how would we recognize
a framework for Consciousness and uh a
related question is what might a
framework for Consciousness be what if
it's a poem Oh Well it can't be a poem
how do we know or what if it's a drawing
or what if it's a set of ontologies
narratives formal documents and tools
could it be that on nft could it be
bolts business operational legal
technical social um could it be
acceptance could it be fundamental
agnosticism um could it be free energy
principle how will we know and that is
really the key question for theories of
Consciousness is not do you feel right
it's how will we know and do you hold
yourself to that standard because we can
do a lot better in this area and this
paper is an example of great work in the
space our next question is what might a
framework for Consciousness enable and
that's pretty broad what would it change
in our action practice would we put more
people in fmri Scanners or less would we
communicate with people who we don't
think of as being able to communicate
with at this point or not why not um how
might it change our legal systems or our
uh social systems what if we could
measure different kinds or types or
levels of consciousness and then that's

really why we need to get down to the
Brass tax on what it might look like
what it measures and what the
predictions
are because if we don't know the unique
predictions or experiments and this is
alluding to earlier when I said that
there's a lot on the table someone's
going to say well I have a Consciousness
Omer and this is a two and this is an 11
and um this type of object that you care
so much about is actually it's not
conscious so I'm just going to unplug it
but this other one that I care a lot
about I measured it with my device and
it was very conscious it's very
worthwhile to save and where will people
make a stand and the stand has to be on
the common ground and the space of
recognition of Frameworks for
Consciousness in my opinion I'm open to
changing my mind on that but I think if
we don't meet at how we'll recognize
such a framework what it might look like
and start with respecting World
Knowledge traditions and by respecting
differences in our opinion and we stay
in a hegemonic mode in a um absolutist
mode in a um non pluralist frame
culturally methodologically if we don't
go down um the pluralistic route it's
going to be I think very bad for
practical
implications and then lastly um what are
the next steps for active inference in
this domain what do we want in active
inference conscious agent to do or how
can we build on active inference now do

we want to go deeper into Consciousness
deeper into action um we spoke about
action policy selection and machine
learning could we put Consciousness in
machine learning what would be the
philosophy and the implications of that
these are all big questions but how are
we going to take this paper and develop
in the directions of active inference
even if only making um educational
content and then lastly um maybe always
a good question to address is what is
the goal of This research sure we're
curious about Consciousness but aren't
we curious about so many things what
makes it worthwhile to do this research
why um are we doing it how have we
selected our how why have we selected
our how how have we selected our why
every one of these combinations we
really need to keep in mind especially
if it's about us about our experience
about Consciousness about people it's
just again can't be said enough that
there's too much on the table to not
have this be participatory inclusive
crosscutting Intercultural the list just
goes on so everybody who's listening and
thinking about it thanks so much this
kind of uh I guess went a little bit
longer than one might have predicted but
say right when active streaming why not
thanks for participating we will provide
follow-up forms to the live participant
we actually will believe it or not we
will um request your feedback
suggestions and questions so if you have
any thoughts again I tried to drop a

bunch of times really specific moments where somebody could just pause the video and say yes this thing right here I do or don't agree with or this does or doesn't resonate with my experience that would be epic because let's hear from a lot of perspectives on Consciousness it's important to hear about that and then also just any feedbacks or suggestions or collaboration offers for the ACT INF stream always welcome you can stay in communication with us any of our channels or means but thank you all for listening those of you who were listening live or those of you who will listen in replay because the future is always Now isn't it see you later